Francisco Zacarias

+351 965437730 | franciscobelchiorzacarias@gmail.com
linkedin.com/in/franciscozacarias | github.com/FranciscoZacarias | franciscozacarias.github.io | NVIDIA Developer Forums

About me

Software engineer focused on low level and systems programming, with experience ranging from drone firmware and Linux kernel drivers to gaming platforms (PC and consoles) and developer tooling. Passionate about performance, 3D, simulation and building systems from scratch. I strong problem solving mindset, I enjoy looking deep and working hard.

Experience

Programmer at Flying Rat August 2026 – Present
Game studio *Prague, Czech Republic*

- Ported game to PS5 and Xbox Series X (Unreal Engine, C++).
- Direct client communication and technical support.
- Multiplayer and online systems (EOS, Steam), networking and platform integration.

Systems and Tools Programmer at GRIP Studios February 2025 – August 2026
Game studio *Prague, Czech Republic*

- Unannounced title: systems programming on custom engine, console ports for Xbox Series X and PS5.
- Unannounced title: UE5 tools programming (city builder plugin, editor utilities, designer support).
- **ICARUS**: UE4 porting and bug fixing for Xbox console.
- **Tomb Raider: Legacy of Atlantis**: UE5 gameplay (quick-time event system) and bug fixing.

Tools Programmer at Toadman Interactive November 2023 – January 2025
Game studio *Berlin, Germany*

- Python backend with Jenkins, Perforce, Unreal Engine and Slack integrations.
- UE bug reporting tool with Jira/Perforce integration, screenshots and context data.
- UE5 crash reporter: parse minidumps/PDBs and auto-symbolicate call stacks across build types.

Software Engineer at Beyond Vision August 2021 – October 2023
R&D drone developer and manufacturer *Lisbon, Portugal*

- C firmware for drone systems: data logging, motherboard routines, retracting gear, fast MAVLink.
- Implemented and debugged Linux kernel drivers for custom hardware.
- ROS (C++/Python) modules for real-time flight data processing and sensor integration.
- Full stack across MAVLink/PX4/ArduPilot, web, Android, databases and multimedia (Janus, GStreamer).

Personal Projects

- **Custom C Standard Library** (GitHub): from-scratch library: math, memory allocators, strings, platform layer.
- **Immediate-mode UI** in C (GitHub) and Jai (GitHub) for my own tools and projects.
- **OpenGL & C 3D Renderer** (GitHub): my first from-scratch 3D prototype engine.
- **C Code Generator** (GitHub): table-driven generation of structured text, replacing compile-time generation.
- **Maze Game** (GitHub): OpenGL/C game built in one week for a game jam.

AI-Assisted Development

I use AI tools daily to iterate, review and debug code faster, and to get up to speed on unfamiliar codebases and APIs. I treat AI output as a draft to review: **design decisions**, **correctness** and **ownership** of the code **stay with me**. Used this way, AI has substantially increased my productivity as a software engineer.

Skills

Languages: C, C++, Python, Rust, Odin, Jai
Embedded & Robotics: Linux kernel drivers, ROS, MAVLink, PX4, ArduPilot, QGroundControl
3D & Graphics: OpenGL, Vulkan, custom engines, Unreal Engine (UE4/UE5)
Platforms & Online: PlayStation, Xbox, Epic Online Services, Steam, networking
Tools & Pipeline: Jenkins, Perforce, Jira, Git